

MLFF SIZE-HALVE2 Fixed-Eight Successive-Fidelity Specification

Coverage-qualified-only q-to-4-to-2-to-1 control plane

mdstats project

2026-08-16

Status

Gate: SIZE-HALVE2

Release: mdstats 0.20.206a0

Architecture revision: 73

Dependency-graph schema: 55

Status: implemented pre-migration control plane

Next gate: SIZE-FIDELITY2

Fixed candidate population

The only possible candidate sizes are 128, 256, 512, 1024, 2048, 4096, 8192, 16384. Dynamic rescue sizes are forbidden. Each candidate binds all REPAIR1 domain-rung identities and its independently rescored MVQUAL1 hard-coverage state.

Hard coverage precedes training. Let q be the count of independently hard-qualified fixed sizes. $q < 4$ fails closed for the future MV funnel. Coverage-failing or unavailable sizes cannot purchase TRAIN2 and cannot be inserted to fill the bracket. MVQUAL1 same-N and N95 non-regression must pass before work is authorized.

Successive-fidelity geometry

The exact survivor geometry is:

3 epochs: $q \rightarrow \min(q, 4)$

10 epochs: $4 \rightarrow 2$

30 epochs: $2 \rightarrow 1$

Every hard-qualified candidate starts once under the common nominal 30-epoch schedule. The 10-epoch run must continue the exact epoch-3 checkpoint, optimizer, and RNG state; the 30-epoch run must continue the exact epoch-10 state. Foundation, evaluation-role, TRAIN2-policy, training-run, and schedule identities cannot change. Normalized schedule progress is exactly 3/30 and 10/30 at the early endpoints, and optimizer-update/structure-exposure counts must strictly increase across each continuation. PERF-P2R stage plans therefore authorize incremental 0→3, 3→10, and 10→30 work without repaying completed epochs.

Ranking and convergence

At epochs 3 and 10, the largest hard-qualified fixed size is preserved only inside its practical-equivalence band. It may still be eliminated when materially worse. Final epoch-30 selection uses the smaller-size preference within practical equivalence together with complete target/replay/physical admissibility. If the largest qualified fixed boundary remains materially better than its smaller admissible finalist, the authority returns **nonconverged_at_fixed_ceiling**.

Migration boundary. `size_half2_plan` is receipt-bound pre-migration evidence, not the production TARGET-DATA2D authority. Revision-64 DATA2C v4/DATA2D v2, DATA8 membership, and generated defaults remain unchanged until SIZE-FIDELITY2 passes and MVMIGRATE1 explicitly migrates them.